

PROJECT: Few-Shot Gaussian Splatting with Diffusion-Based Data Augmentation

Project Overview

This project explores training a Gaussian Splatting (GS) model from a small number of input images. We will augment the limited training set with synthetic views generated by modern diffusion models. The goal is to evaluate how synthetic images affect the final quality of the GS model, with particular attention to whether and how synthetic data degrades performance.

Project Deliverables

- Implement or adapt a GS training pipeline that can run with N real input images (at least 100). Then:
 - Provide a baseline trained only with the N real images.
 - Remove part of the original dataset to end up with a small number of images (5, 10 or 20). Provide one or more augmentation variants: N real images plus S synthetic images produced by diffusion models. Make S configurable so you can test synthetic: real ratios such as 0%, 25%, 50%, 100%, and 200% variants.
- Generate S additional images only from the limited real set using a diffusion model. Recommended strategies:
 - Outpainting or background completion to enlarge the field of view while keeping scene layout.
 - Inpainting to fill occlusions or missing regions.
 - Edge, depth, or pose guided synthesis to approximate nearby viewpoints. Organize outputs by scene with clear filenames linking each synthetic image to its approximate real view.
- Evaluate held out view renderings using PSNR and SSIM metrics (other metric are also welcome if meaningful):
 - Provide perceptual comparisons of renderings vs ground truth with side by side panels.
 - Report training time and memory footprint for each condition.
 - Ablations: compare different diffusion strategies (outpainting vs inpainting vs guided synthesis).

Finally, deliver a clean GitHub repository with training and evaluation scripts, and a README with all the findings.

Report

Write an ~ 5 page report that describes the methodology, augmentation strategies, experiment grid, and results. Include:

- A short description of Gaussian Splatting in the few shots setting.
- A description of the diffusion augmentation methods.
- Tables and plots for the metrics, plus visual examples for each condition.
- A focused discussion on how increasing reliance on synthetic data affects GS performance and when diffusion augmented data is beneficial or harmful.

Final Evaluation Criteria

Rigor of the experimental design: clear baselines, ablations, and controls. Plus:

- Quality and depth of the analysis: strength of the metric results and qualitative insights.
- Reproducibility and packaging: ability to rerun experiments and clarity of organization.
- Presentation: clarity of the report, readability of plots and tables, and effectiveness of the visuals.